

1-2

publish

4

rely

3, 5-7

maintain and improve



no unique repair

1 Learn

analysis optimizer, an LLM agent

question: March store revenue
`SUM(ss_net_paid) ... d_moy = 3`
 13,570,368.70, judged correct ✓
 answer provenance extracted

2 Publish revision 0

after the admission checks

key uniqueness: one row per sale ✓
 date role: d_date_sk unique ✓
 join completeness: loss $\leq$ 1.0% ✓
 evidence: learning SQL, snapshot

3 Optimize: revision 1

period as a date-key range

equal on 14 periods ✓
 29.1 $\rightarrow$ 23.3 ms, -19.8% ✓
 adds date-key contiguity ✓
 revision 0 stays usable

4 User agents rely on it

given only the metric name

September store revenue?
 lookup: revision 1, valid ✓
 declared, run on one snapshot ✓
 13,069,651.80, correct ✓

a later write

5 Restatement

old sales kept as non-current rows

store_sales 13 $\rightarrow$ 14
 key uniqueness fails ✗
 revision 1 invalid ✗
 September: 14,300,525.64 ✗

6 Bounded repair

filters on low-cardinality columns

ss_is_current='1' ✓
 ss_is_current='0' ✗
 exactly one restores the grain ✓

7 Publish revision 2

regression test as of learning time

March, learning SQL 13,570,368.70
 March, repaired SQL 13,570,368.70
 equal: repair accepted ✓
 September: 12,932,888.04 ✓

✗ No unique repair

e.g., a duplicate load

no filter restores the grain ✗
 or several filters disagree ✗
 definition invalidated ✗
 relearned from new tasks

none or several